Category:

Forensic

Name;

F2: CryptedCoin

Message:

A hacker is suspected of stealing cryptocurrency, but the wallet files and chat logs recovered from his PC leave unanswered questions. Your mission is to recover the wallet's private key and identify the destination address.

Instructions:

1. According to chat_log.txt, wallet.dat is encrypted with XOR, and the encryption key appears to be a combination of the main character of the Matrix and the release date.
2. Analyzing the memory dump with strings can reveal important keywords in plain text.

```
$ strings memory_dump.raw | grep pass
Note: XOR key is SHA-256(passphrase)Zs
passphrase=Neo19990331!
SHA256(passphrase)=8bd747e8a8b44487d29475d8c8bf76f546578e2ae1b8b47f524070281755
d234
```

3. Here is a sample python script for decoding wallet.dat.

```
# -*- coding: utf-8 -*-
import hashlib
passphrase = 'Neo19990331!'
key = hashlib.sha256(passphrase.encode()).digest()
with open('wallet.dat', 'rb') as f:
    blob = f.read()
assert blob.startswith(b'WALLETXOR1¥n'), "header mismatch"
cipher = blob[len(b'WALLETXOR1¥n'):]
plain = bytes([b ^ key[i % len(key)] for i, b in enumerate(cipher)])
print(plain.decode('utf-8'))
```

```
$ python decode.py
{
  "type": "bitcoin_wallet",
  "note": "Decrypt me with the correct passphrase (hero+date+symbol).",
  "address": "bc1qw508d6qejxtdg4y5r3zarvary0c5xw7kygt080",
  "wif": "KxFAKExxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx",
  "hint": "XOR with SHA-256(passphrase) repeated",
  "flag": "CSG_FLAG{crypt0_w4ll3t_forensics}"
}
```